

DEXTER FERNANDES

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github.com/Dexter-Fernandes

SUMMARY

Computer Vision Engineer with 3+ years developing perception and vision algorithms from concept through testing, validation and deployment, spanning detection, tracking, segmentation and LiDAR-based SLAM. Delivered a LiDAR and RGB-D sensor fusion SLAM system in ROS2 for autonomous navigation and inspection mapping, and shipped real-time multi-camera perception on edge hardware (NVIDIA DeepStream, OpenVINO, TensorRT) for industrial safety systems in demanding field conditions. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Adapted YOLOv5 detectors in PyTorch, replacing backbone and neck layers and retraining for rig-specific classes, with Optuna sweeps and Weights & Biases tracking across runs.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 perception on BP drilling rigs, covering camera ingestion, inference and field diagnostics in desert conditions.
- Led on-site validation on BP rig deployments, diagnosing model failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Introduced pytest, unittest and linting into CI for edge inference modules, improving release consistency across hardware targets.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes (VTC7252-7C4IP), raising throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and running incident detection under edge compute constraints.
- Tuned camera and ISP settings to keep models reliable on RGB and IR streams across 24/7 operation, cutting false positives on IR and reducing manual recalibration.
- Implemented optical flow based ego-motion estimation feeding CNN classifiers in PyTorch to infer vehicle state, supporting collision avoidance on moving vehicles.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Fit uncertainty-aware ground-plane homographies per camera with a held-out quality gate (median <0.10m, p90 <0.30m), propagating foot-point pixel uncertainty into projection.
- Built an automated model-promotion gate checking match-rate and mean-IOU against a signed SHA-256 manifest, and wired GPU smoke plus MOT17 integration tests into GitHub Actions CI.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

ROS2 SLAM and Nav2 Simulation

02/2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection.

- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer with robot_localization fusing wheel odometry and IMU.
- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND
Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (roslcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.
- Configured Nav2 in simulation for autonomous waypoint traversal over the generated map, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA
Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer Vision & DL	object detection, image classification, semantic and instance segmentation, object tracking, optical flow, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v26), RT-DETR, PyTorch, TensorFlow, Keras, OpenCV, scikit-image
Camera & Sensors	multi-camera synchronisation, ISP and camera tuning (RGB and IR), LiDAR (Velodyne VLP-16), depth sensing (RealSense D435i), sensor fusion, homography and ground-plane calibration, LiDAR extrinsics calibration
Robotics & SLAM	ROS2 (Robot Operating System 2), roslcpp, rclpy, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, GLIM, Nav2, LiDAR SLAM, ICP (Iterative Closest Point), PCL (Point Cloud Library), Open3D, EKF, robot_localization
Edge Inference	TensorRT, OpenVINO, ONNX, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, latency and throughput profiling, VRAM profiling, Intel VTune, NVIDIA Jetson AGX Xavier, Intel Movidius VPU
Video Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, multi-target multi-camera (MTMC) tracking, ByteTrack, NvDCF, NvSORT
MLOps	MLflow, DVC (Data Version Control), Weights & Biases, Optuna, CVAT, benchmarking harnesses, ablation studies, hypothesis testing, drift tracking, Grad-CAM, TIDE error analysis
Testing & VCS	Git, GitHub Actions (CI/CD), pytest, unittest, gtest, Docker, REST API design, containerised microservices, structured logging, observability
Professional	technical leadership, mentoring, Agile and Scrum, Jira, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE
Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)